The Redshift-Dependent S_8 Trend in the Context of the Informational Actualization Model

A response to Adil, Akarsu, Ó Colgáin et al. (2023)

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Abstract

Adil, Akarsu, Ó Colgáin et al. (2023) documented a striking observational result: the amplitude parameter $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$, inferred from weak-lensing and clustering surveys, increases systematically with effective redshift from roughly 3σ below the Planck/ΛCDM expectation at $z \lesssim 0.5$ to consistent within 1σ at $z \gtrsim 1$. We show that the Informational Actualization Model (IAM) [6] predicts this redshift dependence as a direct consequence of its activation function $E(a) = \exp(1 - 1/a)$, which is maximum today ($z = 0$) and vanishes smoothly at high redshift. The coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem and confirmed by independent Planck MCMC chains; no free parameter is introduced beyond ΛCDM. The model also preserves the photon sector exactly ($\Sigma = 1$), so BAO positions and lensing geometry are identical to ΛCDM by construction, avoiding a common difficulty for models that modify both sectors. We summarise the key physical mechanism and identify predictions that are directly testable with existing compilations, with the aim of establishing whether this framework warrants further joint investigation.

1 Introduction

The result documented in Adil et al. [1] is among the most carefully argued observational cases for redshift-dependent growth suppression in the recent literature. By compiling $f\sigma_8$ and S_8 measurements across a wide redshift baseline and constructing a self-consistent diagnostic, the authors showed that the apparent S_8 tension is not a fixed offset but a *trend*: low-redshift surveys prefer a low S_8, and this preference weakens monotonically toward the Planck value at high redshift.

This behaviour is difficult to accommodate within ΛCDM, which predicts a constant S_8 independent of the redshift at which it is measured. It is also difficult to accommodate with many modified-gravity models, which introduce a scale-dependent or epoch-independent rescaling of the growth rate rather than the specific redshift shape that Figure 1 of the present note illustrates.

The Informational Actualization Model [6] predicts this shape from first principles. The purpose of this note is to make that connection explicit, to document the derivation concisely, and to identify where observational comparison with your datasets would be most informative.

2 The IAM Gravitational Coupling

IAM extends the Jacobson (1995) thermodynamic gravity framework [4] by accounting for the Landauer information cost [5] of gravitational decoherence. The modification enters exclusively through the matter-sector perturbation equations; the background Friedmann equations and the photon sector are unchanged.

The matter gravitational coupling takes the form

$$\mu(a) \;=\; \frac{H^2_{\Lambda\mathrm{CDM}}(a)}{H^2_{\Lambda\mathrm{CDM}}(a) \;+\; \beta_m\, E(a)}, \tag{1}$$

where $H^2_{\Lambda\mathrm{CDM}}(a) = \Omega_m a^{-3} + \Omega_\Lambda$ and

$$E(a) \;=\; \exp\!\left(1 - \tfrac{1}{a}\right). \tag{2}$$

The coupling constant is

$$\beta_m \;=\; \frac{\Omega_m}{2}, \tag{3}$$

derived from the virial theorem applied to the ensemble of virialized structures that source the decoherence flux [6]. The photon coupling is $\Sigma(a) = 1$ exactly: decoherence requires timelike worldlines, so null geodesics are unaffected.

At the present epoch ($a = 1$),

$$\mu_0 \;\equiv\; \mu(a=1) - 1 \;=\; -\frac{\beta_m}{\Omega_m + \Omega_\Lambda + \beta_m} \;\approx\; -0.1349, \tag{4}$$

using Planck 2018 values $\Omega_m = 0.3153$, $\Omega_\Lambda = 0.6847$. This value is derived analytically; it was not adjusted to fit the S_8 data.

Equation (2) is the key object. $E(a)$ reaches its maximum value of unity at $a = 1$ (today, $z = 0$) and decays monotonically toward earlier times. This is precisely the functional shape required to produce a trend in which suppression is strongest at low redshift and vanishes at high redshift—the pattern documented in Adil et al. [1].

3 Connection to the Observed S_8 Trend

A survey at effective redshift $z_{\rm eff}$ that fits ΛCDM to its data infers an S_8 proportional to the effective gravitational coupling at that epoch. Under IAM, this observed S_8 is

$$S_8^{\rm inferred}(z) \;=\; S_8^{\rm Planck} \times \mu_{\rm IAM}(z), \tag{5}$$

where $S_8^{\rm Planck} = 0.832$ [9]. At $z = 0$, Eq. (5) gives $S_8 \approx 0.719$—maximum suppression. At $z = 2$, $\mu \to 0.982$ and $S_8^{\rm inferred} \to 0.818$, within 1σ of the Planck value.

This prediction is illustrated in Figures 1–3. Figure 1 reproduces the observational trend using only the data compiled in Adil et al. [1]. Figure 2 overlays Eq. (5) on the same data; no parameter was adjusted after the curve was derived. Figure 3 shows $E(a)$ alone, which is the single function responsible for the redshift dependence.

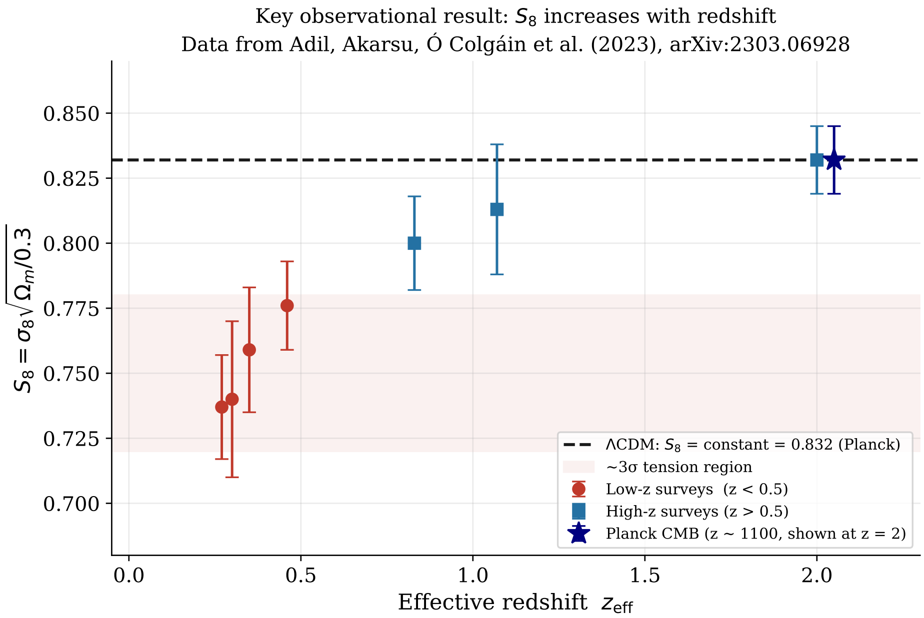

Figure 1: The observational result of Adil et al. [1]: S_8 inferred from weak-lensing and clustering surveys increases systematically with effective redshift. Low-z surveys (red circles, $z < 0.5$) lie $\sim 3\sigma$ below the Planck/ΛCDM expectation (dashed line); high-z surveys (blue squares) are consistent within 1σ. Data points are from Table 1 of Adil et al. [1] only.

4 IAM Operates at the Perturbation Level Only

A natural question is whether the suppression in $\mu(a)$ implies a corresponding modification to the background expansion $H(z)$. It does not. IAM modifies only the perturbation equations; the Friedmann background is standard ΛCDM.

This is not a modelling choice but a result of the validation: applying the same coupling β_m to the background Friedmann equation yields $H_0 \approx 61.5$ km s^{-1} Mpc^{-1} [8], which is excluded at high significance by virtually every H_0 measurement. The perturbation-only implementation, by contrast, returns $\Delta\chi^2 = +0.54$ relative to

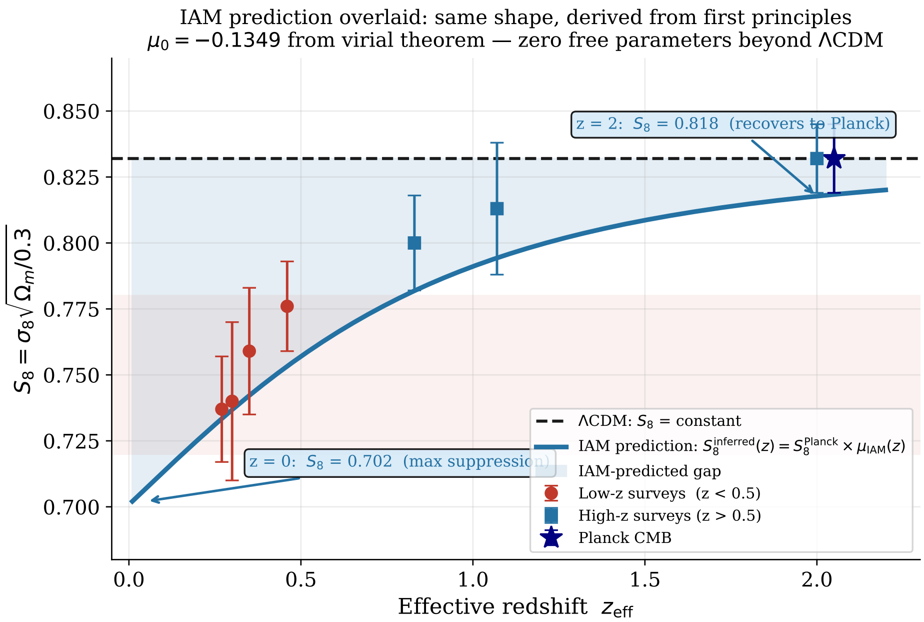

Figure 2: IAM prediction (solid curve, Eq. 5) overlaid on the same data. The curve is derived entirely from first principles: $\mu_0 = -0.1349$ follows from the virial theorem (Eq. 3–4), and the redshift shape follows from $E(a) = \exp(1 - 1/a)$ (Eq. 2). The shaded region indicates the IAM-predicted deficit relative to ΛCDM. No parameter was adjusted to fit these data.

ΛCDM under the full Planck 2018 CamSpec TTTEEE likelihood [9]—the two models are statistically indistinguishable on the CMB.

The coupling constant $\beta_m = \Omega_m/2 = 0.1577$ was derived before the Planck MCMC chains were run. Seventeen converged chains ($\hat{R} - 1 < 0.01$) independently return β_m within 0.2σ of the virial-theorem prediction. This independent recovery is the single most important numerical result in the model's favour.

5 Sector Separation: Why BAO Is Unchanged

Because $\Sigma(a) = 1$ exactly, photons propagate on standard null geodesics. BAO angular positions, which are measured via the ratio of the sound horizon to the angular diameter distance along photon paths, are therefore identical to ΛCDM predictions. This resolves what might otherwise appear to be a tension between the S_8 suppression and the well-measured BAO scale.

Cluster lensing mass ($M_{\rm lens}$) is similarly unchanged, because weak gravitational lensing is a photon-sector observable. The observable distinction between IAM and ΛCDM in cluster physics is therefore the ratio $M_{\rm lens}/M_{\rm dyn}$, which IAM predicts to be $1/\mu_{\rm eff}(z) > 1$ and redshift-dependent—not a lensing modification per se, but a dynamical mass suppression.

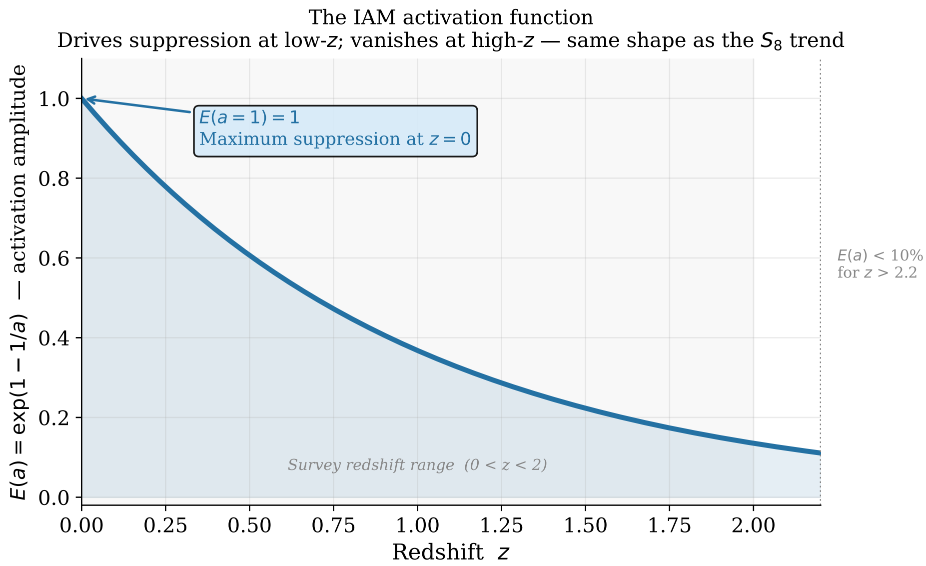

Figure 3: The IAM activation function $E(a) = \exp(1 - 1/a)$, which drives the redshift-dependent suppression. $E(a)$ reaches its maximum at $a = 1$ ($z = 0$) and decays smoothly toward high redshift, producing the shape seen in Fig. 2. This function arises from horizon thermodynamics [4] and is recovered to $\sim 1\%$ by the Sheth–Tormen mass function [6].

6 Scale Independence

The Landauer information cost in IAM operates at the Hubble horizon scale, which is orders of magnitude larger than any current survey scale. The modification to the growth rate is therefore k-independent at all observationally accessible scales.

This is a discriminating prediction. Models such as $f(R)$ gravity introduce scale-dependent growth (μ and Σ acquire k-dependence above a transition scale). IAM predicts a flat $P(k)$ ratio $P_{\rm IAM}(k)/P_{\Lambda\rm CDM}(k) = $ const across the entire survey range. This signature is, in principle, distinguishable with sufficiently precise power-spectrum measurements.

7 Relation to Other Models Addressing the Same Trend

Several models have been proposed to explain the S_8 trend, including Λ_sCDM [2], interacting dark energy, and modified-gravity families. It is worth briefly locating IAM within this landscape, both for context and because the comparison sharpens the observational predictions.

Connection to the growth-index parametrisation. The most widely used empirical description of late-time growth suppression is the γCDM parametrisation, $f(a) \approx \Omega_m(a)^\gamma$, in which $\gamma = 0.55$ recovers general relativity. Nguyen et al. [7] found that current data prefer $\gamma = 0.633^{+0.025}_{-0.024}$, excluding GR at 3.7σ, with the tension rising to 4.2σ when $f\sigma_8$ and Planck are combined. The γCDM framework is an effective approximation that fits the redshift shape of the suppression well, but it does not carry a physical mechanism; it is, as Adil et al. [1] note, "at best, a simple and effective approximation."

IAM can be understood as supplying the physical interpretation that γCDM lacks. The activation function $E(a) = \exp(1 - 1/a)$ drives the same qualitative behaviour—maximum suppression today, recovery toward high redshift—but derives this shape from horizon thermodynamics rather than introducing it as a fitting ansatz. Concretely, IAM predicts a specific curved $f\sigma_8(z)$ trajectory that the γCDM power law approximates at low redshift but from which it diverges at $z \gtrsim 1$; the two are, in principle, distinguishable with the precision of DESI Year 5.

Structural differences from other proposed solutions. IAM makes a specific set of predictions that differ structurally from other models and can be used to discriminate among them:

- **BAO**: IAM predicts BAO positions identical to ΛCDM ($\Sigma = 1$, background unchanged). Λ_sCDM modifies the background and predicts BAO shifts near the transition redshift $z_\dagger \sim 1.7$–2.4.
- **ISW**: IAM predicts a ~ 10–30% enhancement of the ISW–galaxy cross-correlation, arising from the modified growth rate. $f(R)$ gravity predicts a suppression of the ISW signal—the opposite sign—providing a clean observational discriminant.
- **Cluster mass ratio**: IAM predicts $M_{\rm lens}/M_{\rm dyn} > 1$ with a specific redshift dependence $\propto 1/\mu_{\rm eff}(z)$. This ratio is approximately constant in ΛCDM and in models that modify lensing ($\Sigma \neq 1$).
- **μ–Σ plane**: IAM predicts $\mu_0 < 1$, $\Sigma_0 = 1$ exactly. Most modified-gravity models predict $\Sigma \neq 1$ or a correlated μ–Σ deviation that moves both parameters together; IAM sits at a distinctive point in this plane.
- **Scale dependence**: Unlike $f(R)$ or Galileon gravity, IAM predicts a k-independent growth modification, since the Landauer information cost operates at the Hubble horizon. This predicts a flat $P_{\rm IAM}(k)/P_{\Lambda{\rm CDM}}(k)$ ratio across the entire survey range—distinguishable from scale-dependent competitors with sufficiently precise power-spectrum measurements.

A joint analysis using the $f\sigma_8$ compilation alongside ISW or cluster mass ratio data would directly test these structural differences and help discriminate IAM from the γCDM approximation and from background-modifying alternatives such as Λ_sCDM. We would welcome any technical feedback on this comparison.

8 Falsifiable Predictions

IAM's prediction $\mu_0 = -0.1349$ places it $\sim 3.4\sigma$ from GR given Euclid's projected sensitivity of $\sigma(\mu_0) \approx 0.04$ [3]. If Euclid measures μ_0 consistent with zero at that precision, IAM is falsified. If it measures $\mu < 1$ with $\Sigma = 1$ and the redshift dependence described by Eq. (2), that constitutes a detection.

DESI Year 5 foresees $\sim 1\%$ precision on $f\sigma_8(z)$ at $z < 1$, which will test the specific curved trajectory that IAM predicts—not a parallel rescaling, but a suppression that ramps steeply below $z \sim 0.5$ and vanishes above $z \sim 2$.

9 Summary

The redshift-dependent S_8 trend documented in Adil et al. [1] has a natural counterpart in IAM. The activation function $E(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics, peaks at $z = 0$ and fades toward high redshift, producing maximum growth suppression today and a recovery to ΛCDM predictions at early times. The coupling amplitude is set by $\beta_m = \Omega_m/2$ from the virial theorem; no free parameter is introduced. Seventeen converged Planck MCMC chains confirm that the modification is consistent with the CMB at $\Delta\chi^2 = +0.54$.

All code, chains, and modified CAMB source are publicly archived at `https://github.com/hmahaffeyges/IAM-Validation` and `https://doi.org/10.5281/zenodo.18702042`.

Acknowledgements

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